

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12406686
B2
September 02, 2025
Wu; Chih-Wei et al.

Techniques for training a multitask learning model to assess perceived audio quality

Abstract

In various embodiments, a training application trains a multitask learning model to assess perceived audio quality. The training application computes a set of pseudo labels based on a first audio clip and multiple models. The set of pseudo labels specifies metric values for a set of metrics that are relevant to audio quality. The training application also computes a set of feature values for a set of audio features based on the first audio clip. The training application trains a multitask learning model based on the set of feature values and the set of pseudo labels to generate a trained multitask learning model. In operation, the trained multitask learning model maps different sets of feature values for the set of audio features to different sets of predicted labels. Each set of predicted labels specifies estimated metric values for the set of metrics.

Inventors: Wu; Chih-Wei (Los Gatos, CA), Williams; Phillip A. (Los Gatos, CA), Wolcott, IV; William Francis (Los Gatos, CA)

Applicant: NETFLIX, INC. (Los Gatos, CA)

Family ID: 78413054

Assignee: NETFLIX, INC. (Los Gatos, CA)

Appl. No.: 16/905793

Filed: June 18, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210350819 A1	Nov. 11, 2021

Related U.S. Application Data

us-provisional-application US 63021635 20200507

Publication Classification

Int. Cl.: **G10L25/60** (20130101); **G06F17/18** (20060101); **G06F18/2113** (20230101); **G06F18/214** (20230101); **G06N20/00** (20190101); **G10L25/27** (20130101)

U.S. Cl.:

CPC **G10L25/60** (20130101); **G06F18/2113** (20230101); **G06F18/214** (20230101); **G06N20/00** (20190101); **G10L25/27** (20130101); G06F17/18 (20130101)

Field of Classification Search

CPC: G10L (25/60); G10L (25/27); G06N (20/00); G06F (17/18); G06F (18/2113); G06F (18/214); G06K (9/623); G06K (9/6256)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8467893	12/2012	Grancharov	700/94	G10L 25/69
9025780	12/2014	Beerends	704/200	G10L 25/69
9396738	12/2015	Abdelal	N/A	G10L 25/69
9635483	12/2016	Francombe	N/A	H04S 7/301
9870784	12/2017	Sharma	N/A	G10L 25/60
10049674	12/2017	Xiao	N/A	G10L 17/04
10667155	12/2019	Ouyang	N/A	G10L 25/60
10957337	12/2020	Chen	N/A	G10L 17/04
10984818	12/2020	Xiao	N/A	G10L 25/69
2018/0158470	12/2017	Zhu et al.	N/A	N/A
2019/0385480	12/2018	Suzuki	N/A	G10L 15/16
2020/0022007	12/2019	Ouyang et al.	N/A	N/A
2020/0227070	12/2019	Kim et al.	N/A	N/A
2020/0349467	12/2019	Teague	N/A	G06F 16/9027
2020/0402530	12/2019	Güzelarslan	N/A	H04M 3/2236
2021/0125629	12/2020	Bryan	N/A	G10L 25/60
2021/0217403	12/2020	Chae	N/A	N/A
2021/0256988	12/2020	Mauri et al.	N/A	N/A
2021/0264938	12/2020	Yao et al.	N/A	N/A
2021/0272583	12/2020	Liu	N/A	N/A
2021/0350820	12/2020	Wu	N/A	G10L 25/69

OTHER PUBLICATIONS

Kang, Myeongsu, and Jing Tian, "Machine Learning: Data Pre-processing", 2018, Prognostics and Health Management of Electronics: Fundamentals, Machine Learning, and the Internet of Things, Chapter 5, pp. 111-130. (Year: 2018). cited by examiner

Dong, Xuan, and Donald S. Williamson, "An Attention Enhanced Multi-Task Model for Objective Speech Assessment in Real-World Environments", Apr. 2020, ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 911-915. (Year: 2020). cited by examiner

Sharma, Dushyant, Aidan O. T. Hogg, Yu Wang, Amr H. Nour-Eldin, and Patrick A. Naylor, “Non-Intrusive POLQA Estimation of Speech Quality using Recurrent Neural Networks”, 2019, 2019 27th European Signal Processing Conference (EUSIPCO), pp. 1-5. (Year: 2019). cited by examiner

International Search Report for application No. PCT/US2021/030968 dated Jul. 30, 2021. cited by applicant

Choi et al., “Neural MOS prediction for synthesized speech using multi-task learning with spoofing detection and spoofing type classification”, 2021 IEEE spoken language technology workshop (SLT), IEEE, DOI:10.1109/SLT48900.2021.9383533, Jan. 19, 2021, pp. 462-469. cited by applicant

Sharma et al., “Non-Intrusive POLQA Estimation of Speech Quality using Recurrent Neural Networks”, 27th European Signal Processing Conference (EUSIPCO), DOI: 10.23919/EUSIPCO.2019.8902646, Sep. 2, 2019, 5 pages. cited by applicant

Dong et al., “An Attention Enhanced Multi-Task Model for Objective Speech Assessment in Real-World Environments”, IEEE International Conference on Acoustics, Speech and Signal processing (ICASSP), IEEE, DOI: 10.1109/ICASSP40776.2020.9053366, 2020, pp. 911-915. cited by applicant

Avila et al., “Non-Intrusive Speech Quality Assessment Using Neural Networks”, In Proceedings of the International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Mar. 2019, pp. 1-5. cited by applicant

Beerends et al., “Perceptual Objective Listening Quality Assessment (POLQA), The Third Generation ITU-T Standard for End-to-End Speech Quality Measurement Part I—Temporal Alignment”, Journal of the Audio Engineering Society, vol. 61, No. 6, 2013, pp. 366-384. cited by applicant

Biberger et al., “An Objective Audio Quality Measure Based on Power and Envelope Power Cues”, Journal of the Audio Engineering Society, DOI: <https://doi.org/10.17743/jaes.2018.0031>, vol. 66, No. 7/8, Jul./Aug. 2018, pp. 578-593. cited by applicant

Bock et al., “Multi-Task Learning of Tempo and Beat: Learning One To Improve the Other”, In Proceedings of the 20th ISMIR Conference, Nov. 4-8, 2019, pp. 486-493. cited by applicant

Campbell et al., “Audio quality assessment techniques—A review, and recent developments”, Signal Processing, doi:10.1016/j.sigpro.2009.02.015, vol. 89, No. 8, Aug. 2009, pp. 1489-1500. cited by applicant

Caruana, Rich, “Multitask Learning”, Machine learning, vol. 28, No. 35, 1997, pp. 41-75. cited by applicant

Chen et al., “Functional Harmony Recognition of Symbolic Music Data With Multi-Task Recurrent Neural Networks”, In Proceedings of the 19th ISMIR Conference, Sep. 23-27, 2018, pp. 90-97. cited by applicant

Dau et al., “Modeling auditory processing of amplitude modulation. I. Detection and masking with narrow-band carriers”, The Journal of the Acoustical Society of America, vol. 102, No. 5, Nov. 1997, pp. 2892-2905. cited by applicant

Dau et al., “Modeling auditory processing of amplitude modulation. II. Spectral and temporal integration”, The Journal of the Acoustical Society of America, vol. 102, No. 5, Nov. 1997, pp. 2906-2919. cited by applicant

Emiya et al., “Subjective and Objective Quality Assessment of Audio Source Separation”, IEEE Transactions on Audio, Speech, and Language Processing, vol. 19, No. 7, Sep. 2011, pp. 2046-2057. cited by applicant

Hines et al., “VISQOLAudio: An objective audio quality metric for low bitrate codecs”, Journal of the Acoustical Society of America, <http://dx.doi.org/10.1121/1.4921674>, vol. 137, No. 6, Jun. 2015, pp. EL449-EL455. cited by applicant

Hines et al., “VISQOL: an objective speech quality model”, EURASIP Journal on Audio, Speech, and Music Processing, DOI 10.1186/s13636-015-0054-9, vol. 1, No. 13, Dec. 2015, pp. 1-18. cited

by applicant

Huber et al., “PEMO-Q—A New Method for Objective Audio Quality Assessment Using a Model of Auditory Perception”, IEEE Transactions on Audio, Speech and Language Processing, vol. 14, No. 6, Nov. 2006, pp. 1902-1911. cited by applicant

Hung et al., “Multitask Learning for Frame-Level Instrument Recognition”, In Proceedings of the International Conference on Acoustics, Speech, and Signal Processing (ICASSP), arXiv:1811.01143, 2019, 5 pages. cited by applicant

Kates et al., “The Hearing-Aid Speech Quality Index (HASQI) Version 2”, Journal of the Audio Engineering Society, vol. 62, No. 3, Mar. 2014, pp. 99-117. cited by applicant

Kates et al., “The Hearing-Aid Audio Quality Index (Haaqi)”, IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 24, No. 2, Feb. 2016, pp. 354-365. cited by applicant

Manocha et al., “A Differentiable Perceptual Audio Metric Learned from Just Noticeable Differences”, In arXiv—arXiv:2001.04460, May 2020, 6 pages. cited by applicant

Martinez et al., “A No-Reference Audio-Visual Video Quality Metric”, In Proceedings of the European Signal Processing Conference (EUSIPCO), 2014, 5 pages. cited by applicant

Moore, Brian C. J., “Computational models for predicting sound quality”, Acoustical Science and Technology, vol. 41, No. 1, Jan. 2020, pp. 75-82. cited by applicant

Pocta et al., “Subjective and Objective Assessment of Perceived Audio Quality of Current Digital Audio Broadcasting Systems and Web-Casting Applications”, IEEE Transactions on Broadcasting, vol. 61, No. 3, Sep. 2015, pp. 407-415. cited by applicant

Rix et al., “Perceptual Evaluation of Speech Quality (PESQ)—A New Method for Speech Quality Assessment of Telephone Networks and Codecs”, In Proceedings of the International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2001, pp. 749-752. cited by applicant

Rix et al., “Objective Assessment of Speech and Audio Quality—Technology and Applications”, IEEE Transactions on Audio, Speech and Language Processing, DOI 10.1109/TASL.2006.883260, vol. 14, No. 6, Nov. 2006, pp. 1890-1901. cited by applicant

Sloan et al., “Objective Assessment of Perceptual Audio Quality Using VISQOLAudio”, IEEE Transactions on Broadcasting, DOI 10.1109/TBC.2017.2704421, vol. 63, No. 4, Dec. 2017, pp. 693-705. cited by applicant

Thiede et al., “PEAQ—The ITU Standard for Objective Measurement of Perceived Audio Quality”, Journal of the Audio Engineering Society, vol. 48, No. 1, 2000, pp. 3-29. cited by applicant

Torcoli et al., “Comparing the Effect of Audio Coding Artifacts on Objective Quality Measures and on Subjective Ratings”, In Proceedings of the Audio Engineering Society (AES) convention, May 23-26, 2018, pp. 1-10. cited by applicant

International Telecommunication Union, Recommendation ITU-T P.861, “Objective quality measurement of telephone-band (300-3400 Hz) speech codecs”, 1996, 34 pages. cited by applicant

International Telecommunication Union, Recommendation ITU-T P.563:, “Single-ended method for objective speech quality assessment in narrow-band telephony applications”, 2004, 66 pages. cited by applicant

International Telecommunication Union, Recommendation ITU-R BS.1116-3:, “Methods for the subjective assessment of small impairments in audio systems”, 2015, pp. 1-32. cited by applicant

International Telecommunication Union, Recommendation ITU-R BS. 1534-3:, “Method for the subjective assessment of intermediate quality level of audio systems”, 2015, pp. 1-36. cited by applicant

Wang et al., “Image Quality Assessment: From Error Visibility to Structural Similarity”, IEEE Transactions on Image Processing, vol. 13, No. 4, Apr. 2004, pp. 600-612. cited by applicant

Zielinski et al., “On Some Biases Encountered in Modern Audio Quality Listening Tests—A Review”, Journal of the Audio Engineering Society, vol. 56, No. 6, Jun. 2008, pp. 427-451. cited by applicant

TSP Lab Software, “Telecommunications & Signal Processing Laboratory Multimedia Signal Processing”, Retrieved from <http://www-mmsp.ece.mcgill.ca/Documents/Software>, on Jan. 21, 2021, 3 pages. cited by applicant

The PEASS Toolkit—Perceptual Evaluation methods for Audio Source Separation, “The PEASS Software”, Retrieved from <http://bass-db.gforge.inria.fr/peass/PEASS-Software.html>, on Jan. 21, 2021, 2 pages. cited by applicant

Audio and Music, “Quality of Experience Research Lab | QxLab”, Retrieved from <https://qxlab.ucd.ie/index.php/audio-and-music>, on Jan. 14, 2021, 2 pages. cited by applicant

Librosa, Retrieved from <http://librosa.org/doc/latest/index.html>, on Jan. 21, 2021, 3 pages. cited by applicant

GitHub, “alexanderlerch/pyACA”, Retrieved from <https://github.com/alexanderlerch/pyACA>, on Jan. 21, 2021, 4 pages. cited by applicant

GitHub, scikit-learn, “Machine Learning in Python”, Release Highlights for 0.24, Retrieved from <https://scikit-learn.org/stable/>, 2021, 2 pages. cited by applicant

TensorFlow Core v2.4.0, “Module: tf.keras”, Retrieved from https://www.tensorflow.org/api_docs/python/tf/keras, on Jan. 21, 2021, 3 pages. cited by applicant

FFmpeg, “A complete, cross-platform solution to record, convert and stream audio and video”, Retrieved from <https://www.ffmpeg.org/>, on Jan. 21, 2021, 28 pages. cited by applicant

International Telecommunication Union, Recommendation ITU-R BS. 1387-1, Retrieved from <https://www.itu.int/rec/R-REC-BS.1387-1-200111-l/en>, 1998-2001, pp. 1-100. cited by applicant

Results of the public multiformat listening test, Retrieved from <http://listening-test.coresv.net/results.htm>, on Jan. 21, 2021, 49 pages. cited by applicant

Video Quality Databases, Retrieved from <http://www.ene.unb.br/mylene/databases.html>, on Jan. 21, 2021, 6 pages. cited by applicant

GitHub, “Perceptual-Coding-In-Python”, Retrieved from <https://github.com/stephencwelch/Perceptual-Coding-In-Python/tree/master/PEAQPython/PQevalAudioMATLAB>, on Jan. 21, 2021, 2 pages. cited by applicant

GitHub, “Perceptual-Coding-In-Python”, Retrieved from <https://github.com/stephencwelch/Perceptual-Coding-In-Python/tree/master/PEAQPython>, on Jan. 21, 2021, 1 page. cited by applicant

GitHub, “EAQUAL”, Retrieved from <https://github.com/godock/eaqual>, on Jan. 21, 2021, 5 pages. cited by applicant

GitHub, “python-pesq”, Retrieved from <https://github.com/vBaiCai/python-pesq>, on Jan. 21, 2021, 4 pages. cited by applicant

GitHub, “mos-ivr”, Retrieved from <https://github.com/stingerpk/mos-ivr/tree/master/P563>, on Jan. 21, 2021, 5 pages. cited by applicant

GitHub, “peass-software”, Retrieved from https://github.com/CVSSP/peass-software/blob/master/pemo_metric.m, on Jan. 21, 2021, 3 pages. cited by applicant

GitHub, “dmca”, Retrieved from <https://github.com/github/dmca/blob/master/2018/2018-02-14-POLQA.md>, on Jan. 21, 2021, 5 pages. cited by applicant

Non Final Office Action received for U.S. Appl. No. 16/905,810 dated Apr. 1, 2022, 32 pages. cited by applicant

Wilkinghoff et al., “Robust Speaker Identification by Fusing Classification Scores with a Neural Network”, Speech Communication; 13th ITG-Symposium, VDE, Oct. 2018, pp. 261-265. cited by applicant

Zhang et al., “Unsupervised Learning in Cross-Corpus Acoustic Emotion Recognition”, 2011 IEEE Workshop on Automatic Speech Recognition & Understanding, IEEE, 2011, pp. 523-528. cited by applicant

Olsan et al., “Geometric Mean Technique”, Decision Aids for Selection Problems, 1996, pp. 69-80.

cited by applicant

Final Office Action received for U.S. Appl. No. 16/905,810 dated Jul. 20, 2022, 20 pages. cited by applicant

Notice of Allowance received for U.S. Appl. No. 16/905,810 dated Dec. 7, 2022, 14 pages. cited by applicant

Primary Examiner: Washburn; Daniel C

Assistant Examiner: Boggs; James

Attorney, Agent or Firm: Artega Law Group, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority benefit of the U.S. Provisional Patent Application titled, “TECHNIQUES FOR PERCEPTUAL AUDIO QUALITY ASSESSMENT,” filed on May 7, 2020 and having Ser. No. 63/021,635. The subject matter of this related application is hereby incorporated herein by reference.

BACKGROUND

Field of the Various Embodiments

(1) The various embodiments relate generally to computer science, artificial intelligence, and audio technology and, more specifically, to techniques for training a multitask learning model to assess perceived audio quality.

Description of the Related Art

(2) Assessing the quality levels of discrete portions of audio signals known as “audio clips” is an important aspect of developing and verifying certain types of audio algorithms, such as audio coders, filtering algorithms, source separation algorithms, etc. Among other things, applying a given audio algorithm to a representative set of reference audio clips and then determining quality levels of the resulting target audio clips, as perceived by actual listeners, can provide valuable insight into the audio quality versus processing efficiency tradeoffs associated with the audio algorithm or the impact the audio algorithm has on overall perceived audio quality.

(3) Directly determining the perceived audio quality levels of audio clips usually involves conducting a subjective listening test. During a typical subjective listening test, multiple people listen to and manually assign subjective quality scores to a set of training audio clips that represents multiple different types of audio content and various audio distortions or “artifacts.” Because designing and conducting a subjective listening test that yields reliable results can be prohibitively time consuming, some attempts have been made to estimate perceived audio quality levels computationally.

(4) In one approach to computationally estimating the perceived audio quality level of a target audio clip, a software application computes values for a set of audio features based on the target audio clip and the associated source audio clip. Each audio feature characterizes one or more aspects of an audio signal that can impact the sounds that people perceive when listening to the audio signal. The software application then maps the values for the set of audio features to a quality score via a perceptual quality model that has been trained using training data that consists of labeled audio clips. The labels associated with the different audio clips are determined from subjective quality scores assigned by human listeners.

(5) One drawback of the above approach is that the perceptual quality models typically implemented lack robustness. In this regard, because conducting effective subjective listening tests

to determine the different labels for the labeled audio clips can be quite time consuming, perceptual quality models are oftentimes trained using only a small number of labeled audio clips that represent only limited types of audio content and audio artifacts. Consequently, in cases where the target audio clip include types of audio content or types of audio artifacts not represented in any of the labeled audio clips, the perceptual quality model, most likely, is not going to be able to accurately predict how human listeners are going to perceive the quality of those target audio clips. (6) Some other approaches attempt to apply speech-based machine learning models to perceptual audio quality assessment. However, because speech is a narrow subset of audio, speech-based machine learning models are usually unable to accurately predict perceptual audio quality levels for audio clips that include non-verbal sounds, such as music, sound effects, animal noises, etc. (7) As the foregoing illustrates, what is needed in the art are more effective techniques for assessing perceived audio quality levels of audio clips.

SUMMARY

(8) One embodiment of the present invention sets forth a method for training a multitask learning model to assess perceived audio quality. The method includes computing a first set of pseudo labels based on a first audio clip and multiple models, where the first set of pseudo labels specifies metric values for a set of metrics that are relevant to audio quality; computing a first set of feature values for a set of audio features based on the first audio clip; and training the multitask learning model based on the first set of feature values and the first set of pseudo labels to generate a trained multitask learning model, where the trained multitask learning model maps different sets of feature values for the set of audio features to different sets of predicted labels, and where each set of predicted labels specifies estimated metric values for the set of metrics.

(9) At least one technical advantage of the disclosed techniques relative to the prior art is that the disclosed techniques can be used to accurately estimate perceived audio quality levels for audio clips spanning diverse types of audio content and audio distortions. With the disclosed techniques, because the multitask learning model is trained based on unlabeled audio clips, the multitask learning model can be efficiently trained using a broader range of training data relative to prior art techniques. In particular, the training data can represent wide ranges of verbal sounds, non-verbal sounds, and audio distortions. As a result, relative to prior art techniques, the trained multitask learning model can be used to more accurately and reliably assess perceived audio quality levels for audio clips in general. These technical advantages provide one or more technological improvements over prior art approaches.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) So that the manner in which the above recited features of the various embodiments can be understood in detail, a more particular description of the inventive concepts, briefly summarized above, may be had by reference to various embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of the inventive concepts and are therefore not to be considered limiting of scope in any way, and that there are other equally effective embodiments.
- (2) FIG. 1 is a conceptual illustration of a system configured to implement one or more aspects of the various embodiments;
- (3) FIG. 2 is a more detailed illustration of the training application of FIG. 1, according to various embodiments;
- (4) FIG. 3 is a more detailed illustration of the quality inference application of FIG. 1, according to various embodiments;
- (5) FIG. 4 is a flow diagram of method steps for training a multitask learning model to assess

perceived audio quality, according to various embodiments; and

(6) FIG. 5 is a flow diagram of method steps for estimating perceived audio quality using a trained multitask learning model, according to various embodiments.

DETAILED DESCRIPTION

(7) In the following description, numerous specific details are set forth to provide a more thorough understanding of the various embodiments. However, it will be apparent to one skilled in the art that the inventive concepts may be practiced without one or more of these specific details.

(8) Effectively encoding “reference” audio clips is often an important aspect of delivering a variety of media items, such as songs, movies, television shows, etc. After an encoded version of a reference audio clip is received, the encoded version of the reference audio clip is decoded to generate a target audio clip that is played for users. Accordingly, understanding the perceived audio quality levels of target audio clips derived from reference audio clips via an encoder/decoder (“codec”) is useful when developing and/or verifying the audio codec for use in delivering media items. Because manually assessing the perceived audio quality levels of target audio clips can be prohibitive time consuming, some attempts have been made to estimate perceived audio quality levels computationally.

(9) In one approach to computationally estimating the perceived audio quality level of a target audio clip, a software application estimates a quality score using a perceptual quality model that has been trained using labeled audio clips. The labels associated with the different audio clips are usually determined from subjective quality scores assigned by human listeners during subjective listening tests. One drawback of these types of perceptual quality models is that because conducting effective subjective listening tests can be quite time consuming, perceptual quality models are oftentimes trained using only a small number of labeled audio clips that represent only limited types of audio content and audio artifacts. Consequently, the perceptual quality models lack robustness. In particular, when a target audio clip includes types of audio content or types of audio artifacts not represented in any of the labeled audio clips, the perceptual quality model, most likely, is not going to be able to accurately estimate the perceived audio quality level of the target audio clip.

(10) With the disclosed techniques, however, a training application trains a multitask learning (“MTL”) model using a relatively large set of unlabeled training audio clips to jointly predict the quality scores output by multiple perceptual quality models. A quality inference application uses the resulting trained MTL model to compute predicted quality scores for a target audio clip. The quality inference application then aggregates and, optionally, scales the predicted quality scores to generate an overall quality score that estimates the perceived quality level of the audio clip.

(11) In some embodiments, for each of any number of unlabeled training audio clips, the training application generates a pseudo label vector and a training feature vector. Each pseudo label vector includes, without limitation, any number of pseudo labels, where each pseudo label is a scaled version of the quality score for the associated training audio clip as computed using a different perceptual quality model. Each of the training feature vectors includes, without limitation, a set of MTL feature values for the associated unlabeled training audio clip. The set of MTL feature values specifies values for a set of MTL features, where each MTL feature can characterize any aspect of an audio signal in any technically feasible fashion. The training application also computes scaling parameters that map the MTL feature values for different MTL features to a single range. The training application executes a multitask training algorithm on an untrained MTL model based on the training feature vectors and the pseudo label vectors. The training application transmits the resulting trained MTL model and the scaling parameters to the quality inference application.

(12) For each of any number of target audio clips, the quality inference application computes a target feature vector based on the target audio clip, a target reference audio clip, and the scaling parameters. The target feature vector includes, without limitation, a set of MTL feature values for the target audio clip. The quality inference application inputs the target feature vector into the

trained MTL model and, in response, the trained MTL model outputs a quality prediction vector. The quality prediction vector includes, without limitation, a different predicted label for each of the perceptual quality models. Each predicted label predicts a scaled version of the quality score that the associated perceptual quality model would compute for the target audio clip. The quality inference application computes the geometric mean of the predicted labels and then scales the aggregate score to a range typically associated with subjective scores to compute an overall quality score for the target audio clip.

(13) At least one technical advantage of the disclosed techniques relative to the prior art is that the quality inference application can accurately estimate perceived audio quality levels for audio clips spanning diverse types of audio content and audio distortions. Because the training application trains the multitask learning model based on unlabeled data, the multitask learning model can be efficiently trained using a broad range of training data. In particular, the training data can properly represent wide ranges of non-verbal content, verbal content, and audio distortions. As a result, the trained MTL model can accurately and reliably estimate the audio metrics for audio clips in general. These technical advantages provide one or more technological improvements over prior art approaches.

System Overview

(14) FIG. 1 is a conceptual illustration of a system **100** configured to implement one or more aspects of the various embodiments. For explanatory purposes, multiple instances of like objects are denoted with reference numbers identifying the object and parenthetical alphanumeric character(s) identifying the instance where needed. As shown, the system **100** includes, without limitation, compute instances **110(1)** and **110(2)**, an unlabeled dataset **120**, pseudo ground truth models **102(1)-102(M)**, where M is any integer greater than 1, a target audio clip **126**, and a target reference audio clip **128**.

(15) In some alternate embodiments, the system **100** can include, without limitation, any number of compute instances **110**, any number of target audio clips **126**, and any number of target reference audio clips **128**. The components of the system **100** can be distributed across any number of shared geographic locations and/or any number of different geographic locations and/or implemented in one or more cloud computing environments (i.e., encapsulated shared resources, software, data, etc.) in any combination.

(16) As shown, the compute instance **110(1)** includes, without limitation, a processor **112(1)** and a memory **116(1)**, and the compute instance **110(2)** includes, without limitation, a processor **112(2)** and a memory **116(2)**. The compute instances **110(1)** and **110(2)** are also referred to herein individually as “the compute instance **110**” and collectively as “the compute instances **110**.” The processors **112(1)** and **112(2)** are also referred to herein individually as “the processor **112**” and collectively as “the processors **112**.” The memories **116(1)** and **116(2)** are also referred to herein individually as “the memory **116**” and collectively as “the memories **116**.” Each of the compute instances **110** may be implemented in a cloud computing environment, implemented as part of any other distributed computing environment, or implemented in a stand-alone fashion.

(17) The processor **112** may be any instruction execution system, apparatus, or device capable of executing instructions. For example, the processor **112** could comprise a central processing unit, a graphics processing unit, a controller, a micro-controller, a state machine, or any combination thereof. The memory **116** of the compute instance **110** stores content, such as software applications and data, for use by the processor **112** of the compute instance **110**. In some alternate embodiments, each of any number of compute instances **110** may include any number of processors **112** and any number of memories **116** in any combination. In particular, any number of the compute instances **110** (including one) may provide a multiprocessing environment in any technically feasible fashion.

(18) The memory **116** may be one or more of a readily available memory, such as random-access memory, read only memory, floppy disk, hard disk, or any other form of digital storage, local or remote. In some embodiments, a storage (not shown) may supplement or replace the memory **116**.

The storage may include any number and type of external memories that are accessible to the processor **112**. For example, and without limitation, the storage may include a Secure Digital Card, an external Flash memory, a portable compact disc read-only memory, an optical storage device, a magnetic storage device, or any suitable combination of the foregoing.

(19) In general, each of the compute instances **110** is configured to implement one or more applications or subsystems of applications. For explanatory purposes only, each application is described as residing in the memory **116** of a single compute instance **110** and executing on a processor **112** of the single compute instance **110**. However, in alternate embodiments, the functionality of each application may be distributed across any number of other applications that reside in the memories **116** of any number of compute instances **110** and execute on the processors **112** of any number of compute instances **110** in any combination. Further, the functionality of any number of applications or subsystems may be consolidated into a single application or subsystem.

(20) In particular, the compute instance **110(2)** is configured to estimate the perceived audio quality of the target audio clip **126**. As referred to herein, the “perceived” audio quality of a target audio clip **126** is the quality as perceived by listeners of the target audio clip **126**. In some embodiments, the listeners are people. In some alternate embodiments, the listeners can include, without limitation, any number of people, any number and/or types of animals (e.g., dogs), any number and/or types of software applications, and/or any number and/or types of hardware devices in any combination. For example, the listeners could be any number of software applications that automatically transcribe dialog included in the target audio clip **126**.

(21) The target audio clip **126** and the target reference audio clips **128** are both audio clips. As referred to herein, an audio clip is a discrete portion of an audio signal and can include any number and/or types of audio content. For example, an audio clip can include, without limitation, dialogue featuring both male and female voices, sound effects, background music, etc. In some embodiments, the audio clip may be included in a media sequence that also includes any amount and type of video content.

(22) The target audio clip **126** is associated with the target reference audio clip **128**. In some embodiments, the target audio clip **126** is derived from the target reference audio clip **128** via any number and/or types of audio algorithms. Some examples of audio algorithms includes, without limitation, audio coders, filtering algorithms, source separation algorithms, etc. In some embodiments, relative to the target reference audio clip **128**, the target audio clip **126** includes, without limitation, any number of audio distortions. In some embodiments, any number of the audio distortions and artifacts are introduced by one or more audio algorithms. In the same or other embodiments, the target audio clip **126** can include, without limitation, any number and/or types of artifacts.

(23) As described previously herein, one prior art approach to computationally estimating the perceived audio quality level of a target audio clip involves computing a quality score using a conventional perceptual quality model that has been trained using training data that consists of labeled audio clips. The labels associated with the different audio clips are determined from subjective quality scores assigned by human listeners.

(24) One drawback of conventional perceptual quality models trained using labeled audio clips is that these types of perceptual quality models typically lack robustness. More specifically, and as described previously herein, because determining the different labels for the labeled audio clips can be quite time consuming, conventional perceptual quality models are oftentimes trained using only a small number of labeled audio clips that represent only limited types of audio content and audio artifacts. Consequently, in cases where the target audio clip include types of audio content or types of audio artifacts not represented in any of the labeled audio clips, the perceptual quality model, most likely, is not going to be able to accurately predict how human listeners are going to perceive the quality of those target audio clips. Some other conventional approaches to computationally estimating the perceived audio quality level of a target audio clip are targeted toward a subset of

audio content (e.g., speech) and are usually unable to accurately estimate the perceived audio quality level of a target audio clip that include other types of audio content.

Generating a Trained MTL Model Based on Unlabeled Data

(25) To address the above problems, the compute instance **110(1)** includes, without limitation, a training application **130** that generates a trained MTL model **178** based on the unlabeled dataset **120** and the pseudo ground truth models **102(1)-102(M)**. As persons skilled in the art will recognize, training an MTL model involves teaching the MTL model to jointly perform multiple tasks using at least a partially shared representation. To generate the trained MTL model **178**, the training application **130** trains an untrained MTL model (not shown) to jointly perform the M tasks of approximating at least a portion of the functionality of each of the pseudo ground truth models **102(1)-102(M)**. For instance, in some embodiments, the trained MTL model **178** estimates the M outputs of the pseudo ground truth models **102(1)-102(M)**.

(26) In some alternate embodiments, to generate the trained MTL model **178**, the training application **130** trains an untrained MTL model to jointly perform any number and/or types of quality-related assessments. For instance, in some alternate embodiments, the training application **130** trains an untrained MTL model to jointly perform the tasks of predicting the existence of clipping artifact(s) and approximating at least a portion of the functionality of each of the pseudo ground truth models **102(1)-102(M)**. In the same or other embodiments, the trained MTL model **178** estimates a Boolean metric value that corresponds to the existence of clipping artifacts and M perceptual quality metric values that estimate the outputs of the pseudo ground truth models **102(1)-102(M)**.

(27) As shown, the unlabeled dataset **120** includes, without limitation, training audio clips **122(1)-122(N)** and training reference audio clips **124(1)-124(N)**, where N is any positive integer. Each of the training audio clips **122(1)-122(N)** and each of the training reference audio clips **124(1)-124(N)** is an audio clip. The training audio clips **122(1)-122(N)** are associated with the training reference audio clips **124(1)-124(N)**, respectively. In some embodiments, the training audio clips **122(1)-122(N)** are derived from the training reference audio clips **124(1)-124(N)**, respectively, via any number and/or type of audio algorithms. The unlabeled dataset **120** is also referred to herein as “training data.” In some embodiments, the training audio clips **122(1)-122(N)** collectively include a wide range of different types of audio content and numerous types of audio distortions. Consequently, the range of the training data is broad with respect to both types of audio content and types of audio distortions.

(28) In some embodiments, multiple training audio clips **122** can be associated with the same reference audio clip (e.g., derived from the same reference audio clip based on different audio algorithms) and therefore not all of the training reference audio clips **124(1)-124(N)** are unique. Consequently, the total number of unique training audio clips **122** is N, but the total number of unique training reference audio clips **124** can be less than or equal to N.

(29) Each of the pseudo ground truth models **102(1)-102(M)** outputs a value of an associated source metric for an audio clip based on the audio clip and, optionally, an associated reference audio clip. The value of the associated source metric is also referred to herein as an “output value.” In some embodiments, each of the pseudo ground truth models **102(1)-102(M)** encapsulates, without limitation, any number and/or types of discrete portions of software, hardware, firmware and/or data structures, in any combination. For instance, in some embodiments, each of the pseudo ground truth models **102** can be an expert system, a trained neural network, a look-up table, a set of one or more equations, a set of one or more rules, a trained decision tree, any other type of algorithm, any other type of data structure, or any combination thereof. In some alternate embodiments, any number of the pseudo ground truth models **102(1)-102(M)** can be hierarchical models that include, without limitation, any number and/or types of models and/or any number and/or type of algorithms in any combination. In some embodiments, at least one of the pseudo ground truth models **102** is a perceptual quality model that is trained based on subjective scores

assigned by human listeners.

(30) Each of the source metrics can be any type of metric that is relevant to audio quality. As referred to herein, a “metric” can be any defined, quantifiable measure, and a “metric value” for a metric is any type of value (e.g., an integer, a real number, a string, a Boolean, an array of any constituent values, etc.) for the metric that can be determined in any technically feasible fashion. In some embodiments, the source metrics can include, without limitation, any number and type of perceived audio quality metrics, audio quality metrics, metrics that characterize any aspects of an audio signal, metrics that characterize any aspects of audio content included in an audio signal, metrics that characterize any number and/or types of degradations (e.g., artifacts, noise, etc.) associated with an audio signal, etc., in any combination. For explanatory purposes only, the values of the source metrics are also referred to herein as “source metric values.”

(31) In some embodiments, any number of the pseudo ground truth models **102** are intrusive and the remaining pseudo ground truth models **102** are non-intrusive. If the pseudo ground truth model **102** is intrusive, then the pseudo ground truth model **102** determines a source metric value for an audio clip based on both the audio clip and the associated reference audio clip. If the pseudo ground truth model **102** is non-intrusive, then the pseudo ground truth model **102** determines a source metric value for an audio clip based on the audio clip but not the associated reference audio clip.

(32) As described in greater detail in conjunction with FIG. 2, in some embodiments, each of the pseudo ground truth models **102** is a different expert system that computes a value for a different perceived audio quality metric based on an audio clip and the associated reference audio clip. Some other examples of pseudo ground truth models **102** include, without limitation, an equation that computes a signal-to-noise ratio, and a look-up table that returns a bitrate associated with a compressed version of the audio clip, to name a few.

(33) For explanatory purposes only, as used herein, the character ‘x’ denotes an integer from 1 to N, where N is the total number of training audio clips **122**. And, as used herein, the character ‘y’ denotes an integer from 1 to M, where M is the total number of the pseudo ground truth models **102**.

(34) The training application **130** resides in the memory **116(1)** of the compute instance **110(1)** and executes on the processor **112(1)** of the compute instance **110(1)**. As shown, the training application **130** includes, without limitation, a pseudo labeling engine **160**, a pseudo ground truth dataset **162**, a feature extractor **140(1)**, a unit scaling engine **150**, a training feature dataset **152**, and an MTL training engine **170**.

(35) The pseudo labeling engine **160** generates the pseudo ground truth dataset **162** based on the unlabeled dataset **120** and the pseudo ground truth models **102(1)-102(M)**. As shown, the pseudo ground truth dataset **162** includes, without limitation, pseudo label vectors **164(1)-164(N)**. The pseudo label vectors **164(1)-164(N)** are associated with the training audio clips **122(1)-122(N)**, respectively. Although not shown in FIG. 1, each of the pseudo label vectors **164(1)-164(N)** includes, without limitation, M pseudo labels, where each of the pseudo labels is associated with a different pseudo ground truth model **102**. For explanatory purposes only, the y.sup.th pseudo label included in each of the pseudo label vectors **164** is associated with the pseudo ground truth model **102(y)**.

(36) In some embodiments, the y.sup.th pseudo label included in the pseudo label vector **164(x)** is a scaled version of the source metric value generated by the pseudo ground truth model **102(y)** for the training audio clip **122(x)**. To generate the y.sup.th pseudo label that is included in the pseudo label vector **164(x)**, the pseudo labeling engine **160** causes the pseudo ground truth model **102(y)** to compute a source metric value for the training audio clip **122(x)** based on the training audio clip **122(x)** and, optionally, the training reference audio clip **124(x)**. The pseudo labeling engine **160** linearly scales the source metric value to a range of 0 to 1 based on the theoretical range of the associated source metric to compute the y.sup.th pseudo label included in the pseudo label vector

164(x). In some embodiments, the theoretical range of the associated source metric is the theoretical range of an output of the pseudo ground truth model **102(y)**.

(37) As shown, the pseudo ground truth dataset **162** is an input to the MTL training engine **170** that, as described below, generates the trained MTL model **178**. In particular, the pseudo label vector **164(x)** specifies, for the training audio clip **122(x)**, a different value for each of M MTL metrics (not shown). As part of generating the trained MTL model **178**, the MTL training engine **170** associates each of the MTL metrics with a different output of the trained MTL model **178**. Notably, each of the MTL metrics is a scaled version of a different source metric and is therefore associated with a different pseudo ground truth model **102**. Furthermore, each of the MTL metrics is relevant to audio quality.

(38) In alternate embodiments, the pseudo labeling engine **160** can compute the pseudo ground truth dataset **162**, the pseudo label vectors **164**, and the pseudo labels in any technically feasible fashion. In some alternate embodiments, the pseudo labeling engine **160** scales the source metric values to any range using any number and/or types of scaling techniques and based on any amount and/or type of data. In other alternate embodiments, the pseudo labeling engine **160** sets each pseudo label equal to the associated source metric value without performing any scaling operations.

(39) Together, the feature extractor **140(1)** and the unit scaling engine **150** generate the training feature dataset **152** based on the unlabeled dataset **120**. As shown, the training feature dataset **152** includes, without limitation, training feature vectors **154(1)-154(N)**. Although not shown in FIG. 1, each of the training feature vectors **154(1)-154(N)** includes, without limitation, F values for F audio features, where F is any integer greater than one. The F audio features are also referred to herein as “the set of audio features,” and the F values for the F audio features are also referred to herein as “the set of feature values for the set of audio features” and “the set of feature values of the set of audio features.”

(40) As part of generating the trained MTL model **178**, the MTL training engine **170** associates each of the audio features with a different input of the trained MTL model **178**. Each of the audio features is a scaled version of a different source feature, where each source feature is any type of feature that characterizes any aspect of an audio signal in any technically feasible fashion. For instance, in some embodiments, the source features characterize different psycho-acoustics aspects of an audio signal and/or different impacts of the audio signal on the human auditory system. Any number (including zero) of the source features are also features of one or more of the pseudo ground truth models **102**. In some embodiments, each of the source features characterizes the perceptual relevance of an audio signal from a different perspective. The F source features are also referred to herein as “the set of source features,” and the F values for the F source features are also referred to herein as “the set of feature values for the set of source features” and “the set of feature values of the set of source features.”

(41) For each training audio clip **122(x)**, the feature extractor **140(1)** computes F values for the F source features based on the training audio clip **122(x)** and, optionally, the training reference audio clip **124(x)**. The feature extractor **140(1)** may compute the values for the source features in any technically feasible fashion. For explanatory purposes only, a value of a feature is also referred to herein as a “feature value”, a value of an audio feature is also referred to herein as an “audio feature value,” and a value of a source feature is also referred to herein as a “source feature value,”

(42) As shown, the unit scaling engine **150** generate scaling parameters **158** based on the source feature values. Although not shown in FIG. 1, the scaling parameters **158** include, without limitation, any number and type of parameters that are used to scale values for each source feature to values for the corresponding audio feature. Subsequently, the unit scaling engine **150** performs scaling operations on the source feature values based on the scaling parameters **158** to compute audio feature values that are within a shared range (e.g., 0 to 1). The unit scaling engine **150** may generate the scaling parameters **158** and perform any number and type of scaling operations on the source feature values in any technically feasible fashion.

(43) For instance, in some embodiments, the scaling parameters **158** include, without limitation, F min-max parameter pairs (not shown in FIG. 1). Each mix-max parameter pair is associated with a different source feature and specifies the minimum source feature value included in the N associated source feature values and the maximum source feature value included in the N associated source features values. Subsequently, for each of the F source features, the unit scaling engine **150** performs a min-max scaling on the N associated source feature values based on the associated min-max parameter pair to compute the N corresponding audio feature values. Performing min-max scaling on a value is also referred to herein as performing one or more min-max scaling operations on the value.

(44) The unit scaling engine **150** then generates the training feature vectors **154(1)-154(N)** based on the audio feature values that are associated with the training audio clips **122(1)-122(N)**, respectively. The unit scaling engine **150** aggregates the training feature vectors **154(1)-154(N)** to generate the training feature dataset **152**. As shown, the training feature dataset **152** is an input to the MTL training engine **170**. Notably, the training feature vector **154(x)** specifies, for the training audio clip **122(x)**, F values for the F audio features.

(45) In some alternate embodiments, the unit scaling engine **150** can generate the scaling parameters **158** and the training feature dataset **152** in any technically feasible fashion. In some alternate embodiments, the unit scaling engine **150** does not generate the scaling parameters **158**. In the same or other alternate embodiments, the unit scaling engine **150** sets the audio feature values equal to the source feature values instead of scaled source features values.

(46) As shown, the MTL training engine **170** generates the trained MTL model **178** based on the training feature dataset **152** and the pseudo ground truth dataset **162**. In some embodiments, the MTL training engine **170** trains an untrained MTL model (not shown in FIG. 1) to jointly estimate the pseudo label vectors **164(1)-164(N)** based on the training feature vectors **154(1)-154(N)**, respectively. The untrained MTL model can learn multiple tasks in parallel based on at least a partially shared representation.

(47) In some embodiments, the untrained MTL model encapsulates, without limitation, any number and/or types of discrete portions of software, hardware, firmware and/or data structures, in any combination. For example, the untrained MTL model can be a neural network, a decision tree formulated as a multitask problem, a random forest formulated as a multitask problem, etc. In some embodiments, the untrained MTL model can be a hierarchical model that includes any number and/or types of models and/or any number and/or type of algorithms in any combination.

(48) The trained MTL model **178** refers to the final version of the untrained MTL model when the training is complete. In some alternate embodiments, the untrained MTL model is replaced with a pre-trained MTL model, and the trained MTL model **178** refers to the final version of the pre-trained MTL model when the MTL training engine **170** finishes performing training operations on the pre-trained MTL model.

(49) The MTL training engine **170** executes any number and/or types of multitask learning algorithms, other machine learning operations, and/or other operations in any combination to train the untrained MTL model based on the training feature vectors **154(1)-154(N)** and the pseudo label vectors **164(1)-164(N)**. For example, the MTL training engine **170** could execute, without limitation, any number and/or types of multi-label regression algorithms, multi-label classification operations, multitask decision tree algorithms, etc.

(50) As described in greater detail in conjunction with FIG. 2, in some embodiments, the MTL training engine **170** trains a deep neural network having F inputs and M outputs based on optimizing an MTL loss (not shown in FIG. 1). In some embodiments, the inputs to the untrained MTL model are the F training features values included in the training feature vector **154(x)**, and the MTL loss is the mean squared error (“MSE”) between the M outputs of the untrained MTL model and the M pseudo label values included in the pseudo label vector **164(x)**. In some alternate embodiments, the MTL training engine **170** can train any number and/or types of untrained MTL

models using any number and/or types of multitask learning algorithms in any combination.

(51) The F inputs of the untrained MTL model are values for audio features and can be specified in any technically feasible fashion. In some embodiments, while the MTL training engine **170** trains the untrained MTL model, the MTL training engine **170** inputs each of the training feature vectors **154(1)-154(N)** into the untrained MTL model at least once. In some embodiments, inputting the training feature vector **154(x)** into the untrained MTL model refers to inputting each of the F training feature values included in the training feature vector **154(x)** into the untrained MTL model.

(52) The M outputs of the untrained MTL model are estimated values for the MTL metrics and therefore are estimated values for scaled versions of the source metrics. In some embodiments, each of the untrained MTL model and (after training) the trained MTL model **178** outputs a quality prediction vector **186**. As shown, the quality prediction vector **186** includes, without limitation, predicted labels **188(1)-188(M)**. The predicted label **188(y)** is an estimated value for the y.sup.th MTL metric and is associated with the pseudo ground truth model **102(y)**. In some embodiments, outputting the quality prediction vector **186** comprises outputting each of the predicted labels **188**.

(53) After the MTL training engine **170** completes the training of the untrained MTL model, the final version of the untrained MTL model is referred to as the trained MTL model **178**. As shown, the training application **130** provides the trained MTL model **178** and the scaling parameters **158** to the quality inference application **180** and/or any number and/or types of other software applications.

(54) The training application **130** can provide the trained MTL model **178** and the scaling parameters **158** to the quality inference application **180** in any technically feasible fashion. In some embodiments, the training application **130** transmits the trained MTL model **178** and the scaling parameters **158** to the quality inference application **180**. In the same or other embodiments, the training application **130** stores the trained MTL model **178** and the scaling parameters **158** in memory that is accessible to the quality inference application **180**. In some alternate embodiments, the quality inference application **180** acquires (e.g., receives from the training application **130** or reads from memory) the trained MTL model **178** and the scaling parameters **158**.

(55) In some alternate embodiments, the trained MTL model **178** refers to the combination of the untrained MTL model and the final values of any number of parameters (e.g., weights and biases) included in the untrained MTL model. In some alternate embodiments, the training application **130** provides the final values of the parameters included in the untrained MTL model instead of the trained MTL model **178** to the quality inference application **180**.

(56) As shown, the quality inference application **180** computes an overall quality score **198** for the target audio clip **126** based on the target audio clip **126**, the target reference audio clip **128**, the scaling parameters **158**, and the trained MTL model **178**. The overall quality score **198** is also referred to herein as the “audio quality score.” The quality inference application **180** resides in the memory **116(2)** of the compute instance **110(2)** and executes on the processor **112(2)** of the compute instance **110(2)**. As shown, the quality inference application **180** includes, without limitation, a feature extractor **140(2)**, a scaling engine **182**, a target feature vector **184**, the trained MTL model **178**, the quality prediction vector **186**, and an overall quality engine **190**.

(57) The feature extractor **140(2)** is a different instance of the feature extractor **140(1)** described in detail previously herein in the context of the training application **130**. The feature extractor **140(2)** has the same functionality as the feature extractor **140(1)**. The feature extractor **140(2)** computes values for the F source features based on the target audio clip **126** and the target reference audio clip **128**. The feature extractor **140(2)** may compute the values for the source features in any technically feasible fashion.

(58) As shown, the scaling engine **182** generates the target feature vector **184** based on the scaling parameters **158** and the source feature values. The scaling engine **182** can perform any type of scaling that is consistent with the scaling parameters **158** and the unit scaling engine **150**. In some embodiments, and as described previously herein in conjunction with the unit scaling engine **150**,

the scaling parameters **158** includes, without limitation, F min-max parameter pairs (not shown in FIG. **1**). For each of the F source features, the scaling engine **182** performs min-max scaling on the source feature value based on the associated min-max parameter pair to compute the associated audio feature value for the target audio clip **126**.

(59) Subsequently, the scaling engine **182** generates the target feature vector **184**. As described in greater detail in conjunction with FIG. **3**, the target feature vector **184** includes, without limitation, F target feature values (not shown in FIG. **1**). Each of the target feature values is a value for a different audio feature. More specifically, the audio feature values for the target audio clip **126** are also the target feature values. The scaling engine **182** aggregates the audio feature values for the target audio clip **126** to generate the target feature vector **184**.

(60) The quality inference application **180** inputs the target feature vector **184** into the trained MTL model **178** and, in response, the trained MTL model **178** generates the quality prediction vector **186**. As shown, the quality prediction vector **186** includes, without limitation, the predicted labels **188(1)-188(M)**. The predicted label **188(y)** estimates the value of the y.sup.th MTL metric for the target audio clip **126**. In some alternate embodiments, the quality inference application **180** can cause the trained MTL model **178** to generate the quality prediction vector **186** in any technically feasible fashion.

(61) As shown, the overall quality engine **190** generates the overall quality score **198** based on the quality prediction vector **186**. The overall quality score **198** estimates a perceived audio quality level of the target audio clip **126**. In some embodiments, the target audio clip **126** is derived from the target reference audio clip **128** using an audio algorithm, and the overall quality score **198** indicates an audio quality versus processing efficiency tradeoff associated with the audio algorithm and/or a perceptual impact associated with the audio algorithm. Some examples of audio algorithms include, without limitation, audio coders, filtering algorithms, source separation algorithms, etc.

(62) The overall quality engine **190** can compute the overall quality score **198** in any technically feasible fashion. For instance, in some embodiments and as depicted in FIG. **3**, the overall quality engine **190** computes a geometry mean of the predicted labels **188(1)-188(M)** to generate an aggregate score (not shown in FIG. **1**). The overall quality engine **190** linearly scales the aggregate score to a range of 1.0 (lowest) to 5.0 (highest) to generate the overall quality score **198** for the target audio clip **126**. As persons skilled in the art will recognize, the range of 1.0 to 5.0 corresponds to the typical range associated with mean opinion scores (“MOSs”).

(63) In some alternate embodiments, the overall quality engine **190** can perform any number and/or type of aggregation operations, any number and/or type of scaling operations, and/or any number and/or type of any other operations on the predicted labels **188(1)-188(N)** to generate the overall quality score **198**. For instance, in some alternate embodiments, the overall quality engine **190** computes a weighted arithmetic mean of the predicted labels **188(1)-188(M)**, where the weights are determined based on the predominant type of audio content included in the target audio clip **126**. In the same or other embodiments, the overall quality engine **190** filters outliers from the predicted labels **188(1)-188(M)** prior to computing the overall quality score **198**.

(64) In some alternate embodiments, the overall quality engine **190** can be a machine learning model that performs, without limitation any number and/or types of linear and/or non-linear operations based on any number of the predicted labels **188(1)-188(M)** to compute the overall quality score **198**. For example, the overall quality engine **190** could be a linear regression model, a support vector machine, a neural network, a composite model, any other type of model, or any combination thereof.

(65) For explanatory purposes only, the system **100** includes, without limitation, a single target audio clip **126** and a single target reference audio clip **128**, and the quality inference application **180** computes a single overall quality score **198** for the single target audio clip **126**. In alternate embodiments, the system **100** includes, without limitation, any number of target audio clips **126**

and the associated target reference audio clips **128**, and the quality inference application **180** computes a different overall quality score **198** for each of the target audio clips **126**.

(66) Advantageously, because generating the trained MTL model **178** does not rely on obtaining subjective scores from people, the total number of the training audio clips **122** is not constrained by the amount of time required to perform subjective listening tests. Consequently, the types of audio content and artifacts represented in the training audio clips **122** and, therefore, the generality of the trained MTL model **178** can be increased relative to prior art techniques that require labeled training audio clips. Furthermore, because the trained MTL model **178** is trained to perform multiple quality-related assessments jointly, the trained MTL model **178** learns to generalize with respect to the training audio clips **122** based on commonalities and differences between the pseudo ground truth models **102**. As a result, the accuracy and the generality of the trained MTL model **178** are further increased.

(67) Note that the techniques described herein are illustrative rather than restrictive and may be altered without departing from the broader spirit and scope of the invention. Many modifications and variations on the functionality provided by the training application **130**, the pseudo labeling engine **160**, the feature extractors **140(1)-140(2)**, the unit scaling engine **150**, the MTL training engine **170**, the scaling engine **182**, and the overall quality engine **190** will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments

(68) It will be appreciated that the system **100** shown herein is illustrative and that variations and modifications are possible. For instance, in some alternate embodiments, the functionality provided by the unit scaling engine **150** as described herein is integrated into the feature extractor **140(1)**. In the same or other alternate embodiments, the functionality provided by the training application **130** and the functionality provided by the quality inference application **180** are integrated into a single application. Further, the connection topology between the various components in FIG. **1** may be modified as desired.

(69) FIG. **2** is a more detailed illustration of the training application **130** of FIG. **1**, according to various embodiments. In particular, FIG. **2** describes some embodiments in which the total number of pseudo ground truth models **102** (denoted herein as *M*) is equal to four, the total number of training feature values **254** (denoted herein as *F*) included in each of the training feature vectors **154** is equal to four, and the trained MTL model **178** is a trained deep neural network.

(70) Note that the descriptions of the various embodiments have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments.

(71) As shown, the training application **130** generates the trained MTL model **178** based on the unlabeled dataset **120** and the pseudo ground truth models **102(1)-102(4)**. As described previously herein in conjunction with FIG. **1**, the unlabeled dataset **120** includes, without limitation, the training audio clips **122(1)-122(N)** and the training reference audio clips **124(1)-124(N)**. Furthermore, the training audio clip **122(x)** is associated with the training reference audio clip **124(x)**.

(72) Each of the pseudo ground truth models **102(1)-102(4)** is a different expert system that outputs a value of a different source metric for an audio clip based on the audio clip and the associated reference audio clip. As depicted in italics, the pseudo ground truth model **102(1)** is a Hearing-Aid Audio Quality Index (“HAAQI”) expert system that computes values for a metric referred to herein as HAAQI. A value of the HAAQI is also referred to herein as an “HAAQI value.” The pseudo ground truth model **102(2)** is a Perceptual Evaluation of Audio Quality (“PEAQ”) expert system that computes values for a metric referred to herein as PEAQ. A value of the PEAQ is also referred to herein as a “PEAQ value.” The pseudo ground truth model **102(3)** is a Perception Model Quality (“PEMO-Q”) expert system that computes values for a metric referred to herein as PEMO-Q. A value of PEMO-Q is also referred to herein as a “PEMO-Q value.” The pseudo ground truth model

102(4) is a Virtual Speech Quality Objective Listener Audio (“ViSQOLAudio”) expert system that computes values for a metric referred to herein as a ViSQOLAudio metric. A value of the ViSQOLAudio metric is also referred to herein as a “ViSQOLAudio value.”

(73) As shown, the training application **130** includes, without limitation, the pseudo labeling engine **160**, the pseudo ground truth dataset **162**, the feature extractor **140(1)**, the unit scaling engine **150**, the training feature dataset **152**, and the MTL training engine **170**. The pseudo labeling engine **160** generates the pseudo ground truth dataset **162** based on the unlabeled dataset **120** and the pseudo ground truth models **102(1)-102(4)**.

(74) As shown, the pseudo labeling engine **160** includes, without limitation, theoretical ranges **226(1)-226(4)**. The theoretical ranges **226(1)-226(4)** specify the range of source metric values (from a minimum to a maximum) that can, theoretically, be output by the pseudo ground truth models **102(1)-102(4)**, respectively. The theoretical range **226(1)** is associated with the pseudo ground truth model **102(1)** (an HAAQI expert system) and, as depicted in italics, is 0 to 1. The range **226(2)** is associated with the pseudo ground truth model **102(2)** (a PEAQ expert system) and, as depicted in italics, is -4 to 0. The range **226(3)** is associated with the pseudo ground truth model **102(3)** (a PEMO-Q expert system) and, as depicted in italics, is -4 to 0. The range **226(4)** is associated with the pseudo ground truth model **102(4)** (a ViSQOLAudio expert system) and, as depicted in italics, is 1 to 5.

(75) For each training audio clip **122(x)**, the pseudo labeling engine **160** generates a pseudo label vector **164(x)** based on the theoretical ranges **226(1)-226(4)**, the training audio clip **122(x)**, and the training reference audio clip **124(x)**. The pseudo label vector **164(x)** includes, without limitation, pseudo labels **264(x,1)-264(x,4)**. For example, and as shown, the pseudo label vector **164(1)** includes, without limitation, the pseudo labels **264(1,1)-264(1,4)**. As also shown, the pseudo label vector **164(N)** includes, without limitation, the pseudo labels **264(N,1)-264(N,4)**.

(76) To generate the pseudo label vector **164(x)**, the pseudo labeling engine **160** generates the pseudo label **264(x, 1)**, the pseudo label **264(x, 2)**, the pseudo label **264(x, 3)**, and the pseudo label **264(x, 4)**. For explanatory purposes only, the pseudo label **264(x, y)** is associated with both the training audio clip **122(x)** and the pseudo ground truth model **102(y)**. To compute the pseudo label **264(x, y)**, the pseudo labeling engine **160** uses the pseudo ground truth model **102(y)** to compute a value for the associated source metric based on the training audio clip **122(x)** and the training reference audio clip **124(x)**.

(77) The pseudo labeling engine **160** can cause the pseudo ground truth model **102(y)** to compute a value for the associated source metric in any technically feasible fashion. For instance, in some embodiments, the pseudo labeling engine **160** inputs the training audio clip **122(x)** and the training reference audio clip **124(x)** into the pseudo ground truth model **102(y)** and, in response, the pseudo ground truth model **102(y)** outputs a value of the associated source metric for the training audio clip **122(x)**.

(78) The pseudo labeling engine **160** then linearly scales the source metric value to a range of 0 to 1 based on the theoretical range **226(y)** associated with the pseudo ground truth model **102(y)** to generate the scaled source metric value that is associated with both the training audio clip **122(x)** and the pseudo ground truth model **102(y)**. The pseudo labeling engine **160** sets the pseudo label **264(x, y)** equal to the scaled source metric value.

(79) For each of the training audio clips **122(x)**, the pseudo labeling engine **160** aggregates the pseudo labels **264(x, 1)**, **264(x, 2)**, **264(x, 3)**, and **264(x, 4)** to generate the pseudo label vector **164(x)**. Each of the pseudo labels **264(1)-264(N)** therefore includes, without limitation, scaled versions of the source metric values generated by each of the pseudo ground truth models **102(1)-102(4)** for the training audio clips **122(1)-122(N)**, respectively.

(80) As described previously herein, the scaled versions of the source metric values are values of associated audio metrics. Furthermore, the pseudo label vector **164(x)** specifies values of the audio metrics for the training audio clip **122(x)**. For explanatory purposes only, the audio metrics are

denoted herein as a scaled HAAQI, a scaled PEAQ, a scaled PEMO-Q, and a scaled ViSQOLAudio metric. And the pseudo label vector **164(x)** includes, without limitation the pseudo labels **264(x,1)-264(x,4)** that are a scaled HAAQI value, a scaled PEAQ value, a scaled PEMO-Q value, and a scaled ViSQOLAudio metric value, respectively.

(81) For instance, and as depicted, the pseudo label vector **164(1)** includes, without limitation, the pseudo label **264(1,1)** that is a scaled HAAQI value for the training audio clip **122(1)**, the pseudo label **264(1,2)** that is a scaled PEAQ value for the training audio clip **122(1)**, the pseudo label **264(1,3)** that is a scaled PEMO-Q value for the training audio clip **122(1)**, and the pseudo label **264(1,4)** that is a scaled ViSQOLAudio metric value for the training audio clip **122(1)**. The pseudo labeling engine **160** aggregates the pseudo label vectors **164(1)-164(N)** to generate the pseudo ground truth dataset **162**.

(82) As shown, the feature extractor **140(1)** includes, without limitation, feature engines **240(1)-240(4)**. Each of the feature engines **240(1)-240(4)** computes a value of a different source feature for an audio clip based on the audio clip and the associated reference audio clip. As depicted in italics, the source features associated with the feature engines **240(1)-240(4)** are Cepstral Correlation (“CepCorr”), Noise-to-Mask Ratio (“NMR”), Perceptual Similarity Measure (“PSMt”), and Neurogram Similarity Index Measure (“NSIM”), respectively. The feature engines **240(1)-240(4)** compute a CepCorr value, an NMR value, a PSMt value, and an NSIM value, respectively, for an audio clip in any technically feasible fashion. Each of the CepCorr value, the NMR value, the PSMt value, and the NSIM value for an audio clip is a scalar that is proportional to the perceived audio quality level of the audio clip.

(83) CepCorr is a feature that is used by the pseudo ground truth model **102(1)** (an HAAQI expert system). As persons skilled in the art will recognize, HAAQI is targeted towards, without limitation, assessing music quality in audio signals received via hearing aids. NMR is a feature that is used by the pseudo ground truth model **102(2)** (a PEAQ expert system). The NMR value for an audio signal is computed based on the significance of noise with respect to a masking curve. PSMt is a feature that is used by the pseudo ground truth model **102(3)** (a PEMO-Q expert system). The PSMt value for an audio signal is computed based on an effective auditory signal processing process that computes the similarity between the internal representations of the audio signal and the associated audio signal. The internal representations of the audio signal and the associated audio signal are computed via a series of processing steps (e.g., gammatone filterbank, half-wave rectification, modulation filterbank, etc.) that simulate different parts of the auditory system. NSIM is a feature that is used by the pseudo ground truth model **102(4)** (a ViSQOLAudio expert system). NSIM is adapted from an image quality metric known as the Structural Similarity Index that is used to measure the perceived audio quality levels of audio signals.

(84) For each training audio clip **122(x)**, the feature extractor **140(1)** computes a CepCor value, an NMR value, a PSMt value, and an NSIM value for the training audio clip **122(x)** using the feature engines **240(1)-240(4)**, respectively. The feature engines **240(1)-240(4)** compute the Cepcor value, the NRM value, the PSMt value, and the NSIM value, respectively, for the training audio clip **122(x)** based on the training audio clip **122(x)** and the training reference audio clip **124(x)**.

(85) The unit scaling engine **150** generates the scaling parameters **158** and the training feature dataset **152** based on the N CepCor values, the N NMR values, the N PSMt values, and the N NSIM values computed by the feature extractor **140(1)**. As shown, the scaling parameters **158** include, without limitation, min-max parameter pairs **238(1)-238(4)**. Each of the min-max parameter pairs **238** specifies the minimum and the maximum of the associated source feature values. The feature extractor **140(1)** sets the min-max parameter pairs **238(1)-238(4)** equal to the pair of the minimum and the maximum of the N CepCorr values, the N NMR values, the N PSMt values, and the N NSIM values, respectively. Note that the min-max parameter pairs **238(1)-238(4)** delineate the actual ranges of the source feature values that the feature engines **240(1)-240(4)**, respectively, compute for the training audio clips **122(1)-122(N)**.

(86) The unit scaling engine **150** performs min-max scaling on each of the N CepCorr values, the N NMR values, the N PSMt values, and the N NSIM values based on the min-max parameter pairs **238(1)-238(4)**, respectively, to scale each of the source feature values to a range of 0 to 1. Min-max scaling is also commonly referred to as min-max normalization. In this fashion, the unit scaling engine **150** generates N scaled CepCorr values, N scaled NMR values, N scaled PSMt values, and N scaled NSIM values.

(87) For each of the training audio clips **122(x)**, the unit scaling engine **150** generates the training feature vector **154(x)** that includes, without limitation, the training feature values **254(x, 1)**, **254(x, 2)**, **254(x, 3)**, and **254(x, 4)**. More specifically, the unit scaling engine **150** sets the training features values **254(x, 1)**, **254(x, 2)**, **254(x, 3)**, and **254(x, 4)** equal to the scaled CepCorr value, the scaled NMR value, the scaled PSMt value, and the scaled NSIM value, respectively, that are associated with the training audio clip **122(x)**.

(88) As described previously herein in conjunction with FIG. **1**, the scaled versions of the feature values are the audio feature values. More precisely, the training feature vector **154(x)** includes, without limitation, the audio feature values computed by the feature extractor **140(1)** for the training audio clip **122(x)**. In some embodiments, the audio features include, without limitation, a scaled CepCorr, a scaled NMR, a scaled PSMt, and a scaled NSIM. For instance, and as depicted, the training feature vector **154(1)** includes, without limitation, the training feature values **254(1,1)-254(1,4)** that specify a scaled CepCorr value, a scaled NRM value, a scaled PSMt value, and a scaled NSIM value, respectively, for the training audio clip **122(1)**. The unit scaling engine **150** aggregates the training feature vectors **154(1)-154(N)** to generate the training feature dataset **152**.

(89) The MTL training engine **170** generates the trained MTL model **178** based on the training feature dataset **152** and the pseudo ground truth dataset **162**. As shown, in some embodiments, the MTL training engine **170** includes, without limitation, an untrained MTL model **270** and an MTL loss **280**. The untrained MTL model **270** and therefore the trained MTL model **178** has four inputs and four outputs. The four inputs of the untrained MTL model **270** are values for the F audio features and are therefore a scaled CepCorr value, a scaled NMR value, a scaled PSMt value, and a scaled NSIM value.

(90) As described previously herein in conjunction with FIG. **1**, the four outputs of the untrained MTL model **270** are the predicted labels **188(1)-188(4)** that are estimates of the M MTL metrics. In some embodiments, the four outputs are a scaled HAAQI value, a scaled PEAQ value, a scaled PEMO-Q value, and a scaled ViSQOLAudio metric value.

(91) As depicted in *italics*, in some embodiments the untrained MTL model **270** is a deep neural network that is fully-connected and includes, without limitation four hidden layers. Each of the first two hidden layers includes, without limitation, sixty-four neurons, the third hidden layer includes, without limitation, thirty-two neurons, and the fourth hidden layer includes, without limitation, sixteen neurons. Each of the layers uses Rectified Linear Unit functions and is followed by a dropout of 0.3. The output layer includes, without limitation, four neurons that each include, without limitation, a Sigmoid activation function.

(92) In some alternate embodiments, the untrained MTL model **270** is a deep neural network that includes, without limitation, any number of inputs, any number of outputs, any number and/or types of layers that are shared between any number of tasks, and any number and/or types of layers that are not shared between tasks. In some alternate embodiments, the untrained MTL model **270** can encapsulate any number and/or types of discrete portions of software, hardware, firmware and/or data structures, in any combination that, together, can learn multiple tasks in parallel based on at least a partially shared representation. In some alternate embodiments, the untrained MTL model can be a hierarchical model that includes any number and/or types of models and/or any number and/or type of algorithms in any combination that, together, can learn multiple tasks in parallel based on at least a partially shared representation.

(93) The MTL training engine **170** executes any number and/or types of multitask learning algorithms, other machine learning operations, and/or other operations in any combination to train the untrained MTL model **270** based on the pseudo label vectors **164(1)-164(N)** and the training feature vectors **154(1)-154(N)**. As shown in italics, in some embodiments, the MTL training engine **170** trains the untrained MTL model **270** based on optimizing the MTL loss **280**.

(94) When the MTL training engine **170** inputs the training feature vector **154(x)** into the untrained MTL model **270**, the untrained MTL model **270** outputs the quality prediction vector **186**. The MTL training engine **170** sets the MTL loss **280** equal to the MSE between the quality prediction vector **186** and the pseudo label vector **164(x)**. The MTL training engine **170** performs any number (including zero) and/or types of optimization operations on the MTL model **270** based on the MTL loss **280**.

(95) In some alternate embodiments, the untrained MTL model is replaced with a pre-trained MTL model, and the MTL training engine **170** executes any number and/or types of multitask training algorithms on the pre-trained MTL model to generate the trained MTL model **178**. In some alternate embodiments, the MTL training engine **170** executes any number and/or types of multitask training algorithms based on the pseudo label vectors **164(1)-164(N)** and the training feature vectors **154(1)-154(N)** to generate the trained MTL model **178**.

(96) After the MTL training engine **170** completes the training of the untrained MTL model **270**, the final version of the untrained MTL model **270** is referred to as the trained MTL model **178**. The training application **130** provides the trained MTL model **178** and the scaling parameters **158** to the quality inference application **180**. In some alternate embodiments, the training application **130** provides the final values of the parameters (e.g., weights and biases) included in the untrained MTL model **270** instead of the trained MTL model **178** to the quality inference application **180**. In some alternate embodiments, the training application **130** provides any number and/or types of the scaling parameters **158**, trained MTL model **178**, and/or the final values of the parameters (e.g., weights and biases) included in the untrained MTL model **270** in any combination to any number and/or types of software applications.

Estimating Perceived Audio Quality Using a Trained MTL Model

(97) FIG. 3 is a more detailed illustration of the quality inference application **180** of FIG. 1, according to various embodiments. For explanatory purposes only, the quality inference application **180** acquires the scaling parameters **158** and the trained MTL model **178** from the training application **130** of FIG. 2. As shown, the quality inference application **180** includes, without limitation, the feature extractor **140(2)**, the scaling engine **182**, the target feature vector **184**, the trained MTL model **178**, the quality prediction vector **186**, and the overall quality engine **190**.

(98) The feature extractor **140(2)** has the same functionality as the feature extractor **140(1)** described in detail previously herein in conjunction with FIG. 2. As shown, the feature extractor **140(2)** includes, without limitation, the feature engines **240(1)-240(4)**. The feature engines **240(1)-240(4)** compute a CepCorr value, an NMR value, a PSMt value, and an NSIM value, respectively, for the target audio clip **126** based on the target audio clip **126** and the target reference audio clip **128**.

(99) The scaling engine **182** generates the target feature vector **184** based on the scaling parameters **158**, the CepCorr value, the NMR value, the PSMt value, and the NSIM value. As described previously herein in conjunction with FIG. 2, the scaling parameters **158** include, without limitation, the min-max parameter pairs **238(1)-238(4)** that are associated with CepCorr, NMR, PSMt, and NSIM, respectively.

(100) In some embodiments, the scaling engine **182** performs min-max scaling on the CepCorr value based on the min-max parameter pair **238(1)** to scale the CepCorr value to a range of 0 to 1. The scaling engine **182** performs min-max scaling on the NMR value based on the min-max parameter pair **238(2)** to scale the NMR value to a range of 0 to 1. The scaling engine **182** performs min-max scaling on the PSMt value based on the min-max parameter pair **238(3)** to scale the PSMt

value to a range of 0 to 1. The scaling engine **182** performs min-max scaling on the NSIM value based on the min-max parameter pair **238(4)** to scale the NSIM value to a range of 0 to 1.

(101) Subsequently, the scaling engine **182** generates the target feature vector **184** that includes, without limitation, the target feature values **384(1)-384(4)**. As depicted in italics, the scaling engine **182** sets the target feature values **384(1)-384(4)** equal to the scaled CepCorr value, the scaled NMR value, the scaled PSMt value, and the scaled NSIM value, respectively,

(102) The quality inference application **180** inputs the target feature vector **184** into the trained MTL model **178** and, in response, the trained MTL model **178** generates the quality prediction vector **186**. In general, the quality prediction vector **186** estimates values of the MTL metrics for the target audio clip **126**. As shown, the quality prediction vector **186** includes, without limitation, the predicted labels **188(1)-188(4)** that specify, without limitation, a scaled HAAQI value, a scaled PEAQ value, a scaled PEMO-Q value, and a scaled ViSQOLAudio metric value, respectively, for the target audio clip **126**.

(103) The overall quality engine **190** generates the overall quality score **198** based on the quality prediction vector **186**. As shown, the overall quality engine **190** computes aggregate score **392** based on the predicted labels **188(1)-188(4)**. The overall quality engine **190** can compute the aggregate score **392** in any technically feasible fashion. In some embodiments, and as depicted in italics, the overall quality engine **190** sets the aggregate score **392** equal to the geometric mean of the predicted labels **188**. As persons skilled in the art will recognize, setting the aggregate score **392** equal to the geometric mean increases the impact of the lowest predicted label **188** on the aggregate score **392**, thereby accounting for the worst case scenario. The overall quality engine **190** then linearly scales the aggregate score **392** to a range of 1.0 (lowest) to 5.0 to generate the overall quality score **198** for the target audio clip **126**. As persons skilled in the art will recognize, the range of 1.0 to 5.0 is the range of the Absolute Category Rating scale which is often associated with MOSs.

(104) The quality inference application **180** provides the overall quality score **198** to any number and/or types of software applications in any technically feasible fashion. In some embodiments, the quality inference application **180** transmits the overall quality score **198** to any number and/or types of software applications. In the same or other embodiments, the quality inference application **180** stores the overall quality score **198** in memory that is accessible to any number and/or types of software applications.

(105) Advantageously, the overall quality score **198** reflects four different perceptual quality expert systems (an HAAQI expert system, a PEAQ expert system, a PEMO-Q expert system, and a ViSQOLAudio expert system) and audio features derived from different psycho-acoustic principles. Accordingly, empirical results have shown that both the accuracy and the generality of quality inference application **180**, on average, with respect to perceived audio quality is higher than any of the HAAQI, the PEAQ, the PEMO-Q and the ViSQOLAudio expert systems.

(106) FIG. 4 is a flow diagram of method steps for training a multitask learning model to assess perceived audio quality, according to various embodiments. Although the method steps are described with reference to the systems of FIGS. 1-3, persons skilled in the art will understand that any system configured to implement the method steps, in any order, falls within the scope of the present invention.

(107) As shown, a method **400** begins a step **402**, where the pseudo labeling engine **160** uses each of the pseudo ground truth models **102** to compute the values for the associated source metric for the training audio clips **122**. At step **404**, for each of the pseudo ground truth models **102**, the pseudo labeling engine **160** computes the pseudo labels **264** associated with the pseudo ground truth model **102** based on the values for the associated source metric. At step **406**, for each of the training audio clips **122**, the pseudo labeling engine **160** determines the pseudo label vector **164** associated with the training audio clip **122** based on the pseudo labels **264** associated with the training audio clip **122**.

(108) At step **408**, for each of the source features, the feature extractor **140(1)** computes the values of the source feature for the training audio clips **122**. At step **410**, the unit scaling engine **150** computes the scaling parameters **158** for the source features. At step **412**, for each of the training audio clips **122**, the unit scaling engine **150** computes the training feature vector **154** associated with the training audio clip **122** based on the associated values for the source features and the scaling parameters **158**.

(109) At step **414**, the MTL training engine **170** trains the untrained MTL model **270** based on the training feature vectors **154(1)-154(N)** and the pseudo label vectors **164(1)-164(N)** to generate the trained MTL model **178**. At step **416**, the training application **130** provides the trained MTL model **178** and the scaling parameters **158** to the quality inference application **180** and/or any number and/or types of other software applications. The method **400** then terminates.

(110) FIG. 5 is a flow diagram of method steps for estimating perceived audio quality using a trained multitask learning model, according to various embodiments. Although the method steps are described with reference to the systems of FIGS. 1-3, persons skilled in the art will understand that any system configured to implement the method steps, in any order, falls within the scope of the present invention.

(111) As shown, a method **500** begins a step **502**, where the quality inference application **180** acquires the trained MTL model **178** and the scaling parameters **158**. At step **504**, the feature extractor **140(2)** computes the values for the source features for the target audio clip **126**. At step **506**, the scaling engine **182** scales the source features based on the scaling parameters **158** to generate the target feature vector **184**.

(112) At step **508**, the quality inference application **180** inputs the target feature vector **184** into the trained MTL model **178** that, in response, outputs the quality prediction vector **186**. At step **510**, the overall quality engine **190** computes the overall quality score **198** based on the quality prediction vector **186**. At step **512**, the quality inference application **180** transmits the overall quality score **198** to any number and/or types of software applications.

(113) At step **514**, the quality inference application **180** determines whether the quality inference application **180** has received a new target audio clip **126**. If, at step **514**, the quality inference application **180** determines that the quality inference application **180** has not received a new target audio clip **126**, then the method **500** terminates.

(114) If, however, at step **514**, the quality inference application **180** determines that the quality inference application **180** has received a new target audio clip **126**, then the method returns to step **504**, where the feature extractor **140(2)** computes the values for the source features for the new target audio clip **126**. The method **500** continues to cycle through steps **504-514** until the quality inference application **180** determines, at step **514**, that the quality inference application **180** has not received a new target audio clip **126**. The method **500** then terminates.

(115) In sum, the disclosed technique can be used to generate a trained MTL model that is used by a quality inference application to accurately and robustly estimate the perceived audio quality levels of target audio clips. In some embodiments, a training application generates a trained MTL model based on N training audio clips, the training reference audio clips from which the training audio clips are derived, and M pseudo ground truth models, where N and M are integers greater than 1. Each pseudo ground truth model maps an audio clip and, optionally, the associated reference audio clip to a value of a different source metric that is relevant to audio quality.

(116) The training application includes, without limitation, a pseudo labeling engine, a feature extractor, a unit scaling engine, and an MTL training engine. The pseudo labeling engine uses the M pseudo ground truth models to compute a different pseudo label vector for each of the N training audio clips. Each of the pseudo label vectors includes, without limitation, M pseudo labels that specify values for M MTL metrics, respectively. Each of the MTL metrics is relevant to audio quality. For each of the N training audio clips, the feature extractor computes F values for F source features. The unit scaling engine computes scaling parameters based on the values for the source

features. The unit scaling engine computes a different training feature vector for each of the N training audio clips based on the values for the source features and the scaling parameters. The MTL training engine trains an untrained MTL model based on the training feature vectors and the pseudo label vectors. The training application then transmits the trained MTL model and the associated scaling parameters to the quality inference application and/or any number of other software applications.

(117) The quality inference application includes, without limitation, the feature extractor, a scaling engine, the trained MTL model, and an overall quality engine. Upon receiving a target audio clip and an associated target reference audio clip, the feature extractor computes F values for the F source features. The scaling engine scales the F values for the F source features based on the scaling parameters to generate a target feature vector. The quality inference application inputs the target feature vector into the trained MTL model. In response, the trained MTL model outputs a quality prediction vector that includes, without limitation, M predicted labels that estimate values for the M MTL metrics. The overall quality engine computes the geometry mean of the M predicted labels and then scales the geometry mean to generate an overall quality score that estimates the perceived audio quality level of the target audio clip.

(118) At least one technical advantage of the disclosed techniques relative to the prior art is that the quality inference application can use the trained MTL model to reliably and accurately estimate perceived audio quality levels for audio clips spanning diverse types of audio content and audio distortions. Because the training application trains the multitask learning model based on unlabeled data, the multitask learning model can be efficiently trained using a broad range of training data. In particular, the training data can properly represent wide ranges of non-verbal content, verbal content, and audio distortions. As a result, the trained MTL model can accurately and reliably estimate audio metric values for audio clips in general. Furthermore, because the training application trains the MTL model to perform multiple quality-related assessments jointly, the MTL model learns to generalize with respect to the training audio clips based on commonalities and differences between the pseudo ground truth models. Consequently, the accuracy and the generality of the trained MTL model are further increased. These technical advantages provide one or more technological improvements over prior art approaches.

(119) 1. In some embodiments, a computer-implemented method for training a multitask learning model to assess perceived audio quality comprises computing a first plurality of pseudo labels based on a first audio clip and a plurality of models, wherein the first plurality of pseudo labels specifies metric values for a plurality of metrics that are relevant to audio quality, computing a first set of feature values for a set of audio features based on the first audio clip, and training the multitask learning model based on the first set of feature values and the first plurality of pseudo labels to generate a trained multitask learning model, wherein the trained multitask learning model maps different sets of feature values for the set of audio features to different pluralities of predicted labels, and wherein each plurality of predicted labels specifies estimated metric values for the plurality of metrics.

(120) 2. The computer-implemented method of clause 1, wherein a first model included in the plurality of models comprises a Hearing-Aid Audio Quality Index expert system, a Perceptual Evaluation of Audio Quality expert system, a Perception Model Quality expert system, or a Virtual Speech Quality Objective Listener Audio expert system.

(121) 3. The computer-implemented method of clauses 1 or 2, wherein a first audio feature included in the set of audio features is associated with a Cepstral Correlation, a Noise-to-Mask Ratio, a Perceptual Similarity Measure, a Neurogram Similarity Index Measure, or a bitrate.

(122) 4. The computer-implemented method of any of clauses 1-3, wherein computing the first set of feature values comprises computing a plurality of source feature values for a first source feature based on a plurality of audio clips that includes the first audio clip, computing at least one scaling parameter based on the plurality of source feature values, and computing a first feature value for a

first audio feature included in the set of audio features based on a first source feature value included in the plurality of source features values and the at least one scaling parameter, wherein the first feature value is included in the first set of feature values.

(123) 5. The computer-implemented method of any of clauses 1-4, wherein computing the at least one scaling parameter comprises setting a first scaling parameter equal to a minimum source feature value included in the plurality of source feature values, and setting a second scaling parameter equal to a maximum source feature value included in the plurality of source feature values.

(124) 6. The computer-implemented method of any of clauses 1-5, further comprising computing an overall quality score for a second audio clip based on the trained multitask learning model and the at least one scaling parameter.

(125) 7. The computer-implemented method of any of clauses 1-6, wherein computing a first pseudo label included in the first plurality of pseudo labels comprises inputting the first audio clip and a first reference audio clip into a first model included in the plurality of models that, in response, outputs a first metric value for a first metric included in the plurality of metrics, and scaling the first metric value based on a theoretical range of the first metric to generate the first pseudo label.

(126) 8. The computer-implemented method of any of clauses 1-7, wherein training the multitask learning model comprises inputting the first set of feature values into the multitask learning model that, in response, outputs a first plurality of predicted labels, computing a loss based on the first plurality of predicted labels and the first plurality of pseudo labels, and performing one or more optimization operations on the multitask learning model based on the loss.

(127) 9. The computer-implemented method of any of clauses 1-8, wherein computing the loss comprises computing a mean squared error between the first plurality of predicted labels and the first plurality of pseudo labels.

(128) 10. The computer-implemented method of any of clauses 1-9, wherein training the multitask learning model comprises executing a multitask learning algorithm on the multitask learning model based on the first set of feature values, the first plurality of pseudo labels, a second set of feature values associated with a second audio clip, and a second plurality of feature values associated with the second audio clip.

(129) 11. In some embodiments, one or more non-transitory computer readable media include instructions that, when executed by one or more processors, cause the one or more processors to train a multitask learning model to assess perceived audio quality by performing the steps of causing a plurality of models to generate a plurality of output values based on a first audio clip, computing a plurality of pseudo labels based on the plurality of output values, wherein the plurality of pseudo labels specifies metric values for a plurality of metrics that are relevant to audio quality, computing a first set of feature values for a set of audio features based on the first audio clip, and training the multitask learning model based on the first set of feature values and the plurality of pseudo labels to generate a trained multitask learning model, wherein the trained multitask learning model maps different sets of feature values for the set of audio features to different pluralities of predicted labels, and wherein each plurality of predicted labels specifies estimated metric values for the plurality of metrics.

(130) 12. The one or more non-transitory computer readable media of clause 11, wherein a first model included in the plurality of models comprises a perceptual quality model that is trained based on subjective scores assigned by human listeners.

(131) 13. The one or more non-transitory computer readable media of clauses 11 or 12, wherein a first audio feature included in the set of audio features is associated with a Cepstral Correlation, a Noise-to-Mask Ratio, a Perceptual Similarity Measure, a Neurogram Similarity Index Measure, or a bitrate.

(132) 14. The one or more non-transitory computer readable media of any of clauses 11-13,

wherein the multitask learning model comprises at least one of a neural network, a decision tree, or a random forest.

(133) 15. The one or more non-transitory computer readable media of any of clauses 11-14, wherein computing the first set of feature values comprises computing a plurality of source feature values for a first source feature based on a plurality of audio clips that includes the first audio clip, computing at least one scaling parameter based on the plurality of source feature values, and computing a first feature value for a first audio feature included in the set of audio features based on a first source feature value included in the plurality of source features values and the at least one scaling parameter, wherein the first feature value is included in the first set of feature values.

(134) 16. The one or more non-transitory computer readable media of any of clauses 11-15, wherein computing the first feature value comprises performing at least one min-max scaling operation on the first source feature value.

(135) 17. The one or more non-transitory computer readable media of any of clauses 11-16, further comprising computing an overall quality score for a second audio clip based on the trained multitask learning model and the at least one scaling parameter.

(136) 18. The one or more non-transitory computer readable media of any of clauses 11-17, wherein computing a first pseudo label included in the plurality of pseudo labels comprises scaling a first output value included in the plurality of output values based on a theoretical range of a first output of a first model included in the plurality of models to generate the first pseudo label.

(137) 19. The one or more non-transitory computer readable media of any of clauses 11-18, wherein training the multitask learning model comprises inputting the first set of feature values into the multitask learning model that, in response, outputs a first plurality of predicted labels, computing a loss based on the first plurality of predicted labels and the plurality of pseudo labels, and performing one or more optimization operations on the multitask learning model based on the loss.

(138) 20. In some embodiments, a system comprises one or more memories storing instructions and one or more processors coupled to the one or more memories that, when executing the instructions, perform the steps of computing a first plurality of pseudo labels based on a first audio clip and a plurality of models, wherein the first plurality of pseudo labels specifies metric values for a plurality of metrics that are relevant to audio quality, computing a first set of feature values for a set of audio features based on the first audio clip, and executing at least one multitask learning algorithm based on the first set of feature values and the first plurality of pseudo labels to generate a trained multitask learning model, wherein the trained multitask learning model maps different sets of feature values for the set of audio features to different pluralities of predicted labels, and wherein each plurality of predicted labels specifies estimated metric values for the plurality of metrics.

(139) Any and all combinations of any of the claim elements recited in any of the claims and/or any elements described in this application, in any fashion, fall within the contemplated scope of the embodiments and protection.

(140) The descriptions of the various embodiments have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments.

(141) Aspects of the present embodiments may be embodied as a system, method or computer program product. Accordingly, aspects of the present disclosure may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “module,” a “system,” or a “computer.” In addition, any hardware and/or software technique, process, function, component, engine, module, or system described in the present disclosure may be implemented as a circuit or set of circuits. Furthermore,

aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

(142) Any combination of one or more computer readable medium(s) may be utilized. The computer readable medium may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random access memory, a read-only memory, an erasable programmable read-only memory (EPROM or Flash memory), an optical fiber, a portable compact disc read-only memory, an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.

(143) Aspects of the present disclosure are described above with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine. The instructions, when executed via the processor of the computer or other programmable data processing apparatus, enable the implementation of the functions/acts specified in the flowchart and/or block diagram block or blocks. Such processors may be, without limitation, general purpose processors, special-purpose processors, application-specific processors, or field-programmable gate arrays.

(144) The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods and computer program products according to various embodiments of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

(145) While the preceding is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A computer-implemented method for training a multitask learning model to assess perceived audio quality, the method comprising: inputting a first training audio clip to a plurality of audio quality models that are separate from the multitask learning model to compute a first plurality of

pseudo labels, wherein the first training audio clip is derived from a separate training reference audio clip, wherein each pseudo label included in the first plurality of pseudo labels specifies a metric value that is relevant to audio quality as measured by an audio quality model of the plurality of audio quality models; performing one or more first scaling operations to scale each pseudo label included in the first plurality of pseudo labels based on a theoretical range of an output of a corresponding audio quality model included in the plurality of audio quality models to generate a first plurality of scaled pseudo labels; computing a first set of feature values for a set of audio features based on the first training audio clip; performing one or more second scaling operations to scale the first set of feature values using a set of scaling parameters computed based on the first set of feature values to generate a first set of scaled feature values; training the multitask learning model based on the first set of scaled feature values and the first plurality of scaled pseudo labels to generate a trained multitask learning model that estimates the first plurality of scaled pseudo labels based on the first set of scaled feature values; and transmitting the trained multitask learning model along with the set of scaling parameters to at least one compute instance, wherein the at least one compute instance, in operation, extracts a set of target features from a second audio clip, scales the set of target features based on the set of scaling parameters received with the trained multitask learning model to generate a second set of scaled feature values, and inputs the second set of scaled feature values to the trained multitask learning model, wherein the trained multitask learning model, in operation, maps, for the second audio clip, the second set of scaled feature values for the set of audio features based on the second audio clip to a plurality of predicted labels, and wherein the plurality of predicted labels specifies estimated metric values for a plurality of metrics relevant to audio quality that would be computed according to the plurality of audio quality models for the second audio clip.

2. The computer-implemented method of claim 1, wherein a first audio quality model included in the plurality of audio quality models comprises a Hearing-Aid Audio Quality Index expert system, a Perceptual Evaluation of Audio Quality expert system, a Perception Model Quality expert system, or a Virtual Speech Quality Objective Listener Audio expert system.

3. The computer-implemented method of claim 1, wherein a first audio feature included in the set of audio features based on the first training audio clip is associated with a Cepstral Correlation, a Noise-to-Mask Ratio, a Perceptual Similarity Measure, a Neurogram Similarity Index Measure, or a bitrate.

4. The computer-implemented method of claim 1, wherein computing the first set of feature values comprises: computing a plurality of source feature values for a first source feature based on a plurality of training audio clips that includes the first training audio clip; computing at least one scaling parameter included in the set of scaling parameters based on the plurality of source feature values; and computing a first feature value for a first audio feature included in the set of audio features based on a first source feature value included in the plurality of source feature values and the at least one scaling parameter, wherein the first feature value is included in the first set of feature values.

5. The computer-implemented method of claim 4, wherein computing the at least one scaling parameter comprises: setting a first scaling parameter equal to a minimum source feature value included in the plurality of source feature values; and setting a second scaling parameter equal to a maximum source feature value included in the plurality of source feature values.

6. The computer-implemented method of claim 4, further comprising computing an overall quality score for the second audio clip based on the trained multitask learning model and the at least one scaling parameter.

7. The computer-implemented method of claim 1, wherein: computing a first pseudo label included in the first plurality of pseudo labels comprises inputting the first training audio clip and the separate training reference audio clip into a first audio quality model included in the plurality of audio quality models that, in response, outputs a first metric value for a first metric included in the

plurality of metrics, and scaling the first pseudo label to generate a first scaled pseudo label included in the first plurality of scaled pseudo labels comprises scaling the first metric value based on a theoretical minimum and a theoretical maximum of the first metric computed by the first audio quality model.

8. The computer-implemented method of claim 1, wherein training the multitask learning model based on the first set of scaled feature values and the first plurality of scaled pseudo labels comprises: inputting the first set of scaled feature values into the multitask learning model that, in response, outputs a first plurality of predicted labels; computing a loss based on the first plurality of predicted labels and the first plurality of scaled pseudo labels; and performing one or more optimization operations on the multitask learning model based on the loss.

9. The computer-implemented method of claim 8, wherein computing the loss comprises computing a mean squared error between the first plurality of predicted labels and the first plurality of scaled pseudo labels.

10. The computer-implemented method of claim 1, wherein training the multitask learning model based on the first set of scaled feature values and the first plurality of pseudo labels comprises executing a multitask learning algorithm on the multitask learning model based on the first set of scaled feature values, the first plurality of scaled pseudo labels, an additional set of feature values associated with an additional training audio clip, and an additional plurality of scaled feature values associated with the additional training audio clip.

11. One or more non-transitory computer readable media including instructions that, when executed by one or more processors, cause the one or more processors to train a multitask learning model to assess perceived audio quality by performing the steps of: inputting a first training audio clip to a plurality of audio quality models that are separate from the multitask learning model to generate a plurality of output values, wherein the first training audio clip is derived from a separate training reference audio clip; computing a plurality of pseudo labels based on the plurality of output values, wherein each pseudo label included in the plurality of pseudo labels specifies a metric value that is relevant to audio quality as measured by an audio quality model of the plurality of audio quality models; performing one or more first scaling operations to scale each pseudo label included in the plurality of pseudo labels based on a theoretical range of an output of a corresponding audio quality model included in the plurality of audio quality models to generate a plurality of scaled pseudo labels; computing a first set of feature values for a set of audio features based on the first training audio clip; performing one or more second scaling operations to scale the first set of feature values using a set of scaling parameters computed based on the first set of feature values to generate a first set of scaled feature values; training the multitask learning model based on the first set of scaled feature values and the plurality of scaled pseudo labels to generate a trained multitask learning model that estimates the plurality of scaled pseudo labels based on the first set of scaled feature values; and transmitting the trained multitask learning model along with the set of scaling parameters to at least one compute instance, wherein the at least one compute instance, in operation, extracts a set of target features from a second audio clip, scales the set of target features based on the set of scaling parameters received with the trained multitask learning model to generate a second set of scaled feature values, and inputs the second set of scaled feature values to the trained multitask learning model, wherein the trained multitask learning model, in operation, maps, for the second audio clip, the second set of scaled feature values for the set of audio features based on the second audio clip to a plurality of predicted labels, and wherein the plurality of predicted labels specifies estimated metric values for a plurality of metrics relevant to audio quality that would be computed according to the plurality of audio quality models for the second audio clip.

12. The one or more non-transitory computer readable media of claim 11, wherein a first audio quality model included in the plurality of audio quality models comprises a perceptual quality model that is trained based on subjective scores assigned by human listeners.

13. The one or more non-transitory computer readable media of claim 11, wherein a first audio feature included in the set of audio features is associated with a Cepstral Correlation, a Noise-to-Mask Ratio, a Perceptual Similarity Measure, a Neurogram Similarity Index Measure, or a bitrate.

14. The one or more non-transitory computer readable media of claim 11, wherein the multitask learning model comprises at least one of a neural network, a decision tree, or a random forest.

15. The one or more non-transitory computer readable media of claim 11, wherein computing the first set of feature values comprises: computing a plurality of source feature values for a first source feature based on a plurality of training audio clips that includes the first training audio clip; computing at least one scaling parameter included in the set of scaling parameters based on the plurality of source feature values; and computing a first feature value for a first audio feature included in the set of audio features based on a first source feature value included in the plurality of source feature values and the at least one scaling parameter, wherein the first feature value is included in the first set of feature values.

16. The one or more non-transitory computer readable media of claim 15, wherein computing the first feature value comprises performing at least one min-max scaling operation on the first source feature value.

17. The one or more non-transitory computer readable media of claim 15, further comprising computing an overall quality score for the second audio clip based on the trained multitask learning model and the at least one scaling parameter.

18. The one or more non-transitory computer readable media of claim 11, wherein a first pseudo label included in the plurality of pseudo labels is scaled based on a theoretical minimum and a theoretical maximum of a first output of a first audio quality model included in the plurality of audio quality models.

19. The one or more non-transitory computer readable media of claim 11, wherein training the multitask learning model based on the first set of scaled feature values and the plurality of scaled pseudo labels comprises: inputting the first set of scaled feature values into the multitask learning model that, in response, outputs a first plurality of predicted labels; computing a loss based on the first plurality of predicted labels and the plurality of scaled pseudo labels; and performing one or more optimization operations on the multitask learning model based on the loss.

20. A system comprising: one or more memories storing instructions; and one or more processors coupled to the one or more memories that, when executing the instructions, perform the steps of: inputting a first training audio clip to a plurality of audio quality models that are separate from the multitask learning model to compute a first plurality of pseudo labels, wherein the first training audio clip is derived from a separate training reference audio clip, wherein each pseudo label included in the first plurality of pseudo labels specifies a metric value that is relevant to audio quality as measured by an audio quality model of the plurality of audio quality models; performing one or more first scaling operations to scale each pseudo label included in the first plurality of pseudo labels based on a theoretical range of an output of a corresponding audio quality model included in the plurality of audio quality models to generate a first plurality of scaled pseudo labels; computing a first set of feature values for a set of audio features based on the first training audio clip; performing one or more second scaling operations to scale the first set of feature values using a set of scaling parameters computed based on the first set of feature values to generate a first set of scaled feature values; executing at least one multitask learning algorithm based on the first set of scaled feature values and the first plurality of scaled pseudo labels to generate a trained multitask learning model that estimates the first plurality of scaled pseudo labels based on the first set of scaled feature values; and transmitting the trained multitask learning model along with the set of scaling parameters to at least one compute instance, wherein the at least one compute instance, in operation, extracts a set of target features from a second audio clip, scales the set of target features based on the set of scaling parameters received with the trained multitask learning model to generate a second set of scaled feature values, and inputs the second set of scaled feature values to

the trained multitask learning model, wherein the trained multitask learning model, in operation, maps, for the second audio clip, the second set of scaled feature values for the set of audio features based on the second audio clip to a plurality of predicted labels, and wherein the plurality of predicted labels specifies estimated metric values for a plurality of metrics relevant to audio quality that would be computed according to the plurality of audio quality models for the second audio clip.
